

Summer Internship Report
on

Reproducing Spin-Direction Coupling in Plasmonic Crystals using Numerical Simulation

Submitted by

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Declaration

I, Ritik Dubey, declare that the work presented in this report is my own and was completed during my summer internship. Any information or content taken from other sources has been properly cited.

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Ritik Dubey
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– Ritik Dubey

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1. Introduction

This summer internship report investigates the fascinating interplay between light's spin and its propagation direction in waveguided plasmonic crystals (WPCs), a phenomenon termed spin-direction-spin coupling, rooted in photonic spin-orbit interactions (SOI). WPCs, consisting of periodic gold gratings on an indium tin oxide waveguide layer, enable the study of two key effects: spin-dependent directional guiding, where light's polarization influences its path, and direction-dependent spin acquisition, where the propagation direction alters the polarization state. Using a custom-designed dark-field Fourier domain Mueller matrix microscopy system, we aim to observe and quantify these effects, which are significantly enhanced by Fano resonance due to mode hybridization.

The objective is to demonstrate the potential of these interactions for developing advanced nanophotonic devices, such as optical switches, polarization sorters, and sensors, which could transform fields like optical communication and nanotechnology. Looking forward, this research opens avenues for applications in quantum optics and telecommunications, offering new methods to manipulate light's properties at the nanoscale and potentially extending these techniques to other nanostructured materials for broader scientific impact.

2. Experimental setup

2.1 Overview

The experimental setup is a custom-designed darkfield Fourier domain Mueller matrix imaging and spectroscopy system, integrated with an inverted microscope. This setup is tailored to investigate spin-direction-spin coupling in waveguided plasmonic crystals (WPCs) by analyzing the polarization and momentum of scattered light in the far field, enabling the study of photonic spin-orbit interactions (SOI).

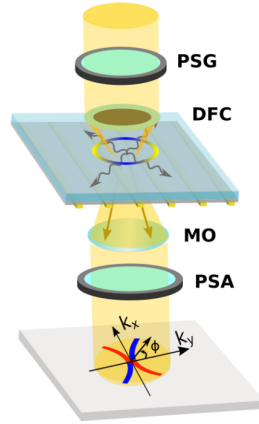


Figure 1: A schematic of the darkfield Fourier domain polarization Mueller matrix microscopy system.

2.2 The Setup

The setup can be visualized as a sequence of optical components arranged to manipulate and analyze light:

- **Light Sources:** A broadband white light source (400–725 nm) and a 405-nm diode laser provide illumination to excite plasmonic and waveguide modes.
- **Polarization State Generator (PSG):** Comprises a rotatable linear polarizer and an achromatic quarter-wave plate to generate six polarization states (horizontal, vertical, $\pm 45^\circ$, left/right circular).
- **Darkfield Condenser (DFC):** A high numerical aperture ($NA \approx 0.92$) condenser focuses light into an annular shape at the WPC sample, ensuring only scattered light is collected to minimize background noise.
- **Sample (WPC):** The WPC is placed at the focal point of the DFC for interaction with the incident light.
- **Microscope Objective:** An Olympus MPlanFL objective ($NA\ 0.5$) collects the scattered light for analysis.
- **Polarization State Analyzer (PSA):** Includes a rotatable QWP and linear polarizer to analyze the polarization state of scattered light.

- **Detection System:** Scattered light is directed to a spectrometer for spectral analysis or an EM CCD camera for imaging, with a Fourier transforming lens ($f = 75$ mm) for momentum domain analysis.

All polarizing elements are mounted on motorized rotational mounts for precise control.

2.3 Procedure for Taking Measurements

The measurement process involves exciting the WPC sample with a broadband white light source spanning 400–725 nm, which can be unpolarized or circularly polarized (right or left) to probe various aspects of spin-orbit interactions. The Polarization State Generator (PSG) produces six distinct polarization states, including horizontal, vertical, $\pm 45^\circ$, and right/left circular polarizations. For each input polarization state, the scattered light from the WPC is analyzed by the Polarization State Analyzer (PSA), which measures the intensity for corresponding output polarization states, resulting in 36 polarization-resolved intensity measurements. These measurements are used to construct the 4x4 Mueller matrix, which comprehensively describes the sample's polarization transformations, capturing effects like diattenuation and retardation. Measurements are conducted at various wavelengths within the broadband range to observe spectral variations, while Fourier domain images are recorded to analyze the momentum-space distribution of the scattered light, revealing wavevector-dependent spin acquisition and directional guiding effects. The Mueller matrix elements are then analyzed to quantify and interpret the spin-direction-spin coupling phenomena, including the forward and inverse spin Hall effects.

HH+HV+VH+VV	HH+HV−VH−VV	PH+PV−MH−MV	RH+RV−LH−LV
HH−HV+VH−VV	HH−HV−VH+VV	PH−PV−MH+MV	RH−RV−LH+LV
HP+VP−HM−VM	HP−VP−HM+VM	PP−PM−MP+MM	RP−RM−LP+LM
HR+VR−HL−VL	HR−VR−HL+VL	PR−PR−MR+ML	RR−RL−LR+LL

Table 1 | Scheme for construction of Mueller matrix using 36 polarization-resolved projective measurements. Here, the first letter represents the input polarization state, the second letter stands for the analyzer or the projected polarization state. The states are defined as I_H (horizontal), I_V (vertical), I_P (+45 deg), I_M (−45 deg), I_L left circular polarized (LCP), I_R (RCP)

2.4 Material: Waveguided Plasmonic Crystal (WPC)

The WPC is a nanostructured metamaterial designed to exhibit hybridized surface plasmon and waveguide resonances:

- **Structure:** Consists of periodic gold (Au) gratings (periodicity ≈ 550 nm, width ≈ 90 nm, thickness ≈ 20 nm) on a 110 nm indium tin oxide (ITO) waveguide layer, deposited on a glass substrate.
- **Fabrication:** Fabricated using electron beam lithography and thermal evaporation, with the ITO layer thickness optimized for spectral overlap between plasmonic and waveguide modes in the 420–600 nm range.
- **Function:** The WPC supports leaky quasiguided modes, enabling the observation of spin-orbit interactions, such as forward and inverse spin Hall effects, and enhancing Fano resonances that amplify the polarization-dependent effects observed in the experiments.

3. Theoretical Background

3.1 Role of DFC in Spin-Orbit Interaction

The Dark-Field Condenser (DFC) is a critical component in the experimental setup for studying spin-direction-spin coupling in waveguided plasmonic crystals (WPCs). Its high numerical aperture ($\text{NA} \approx 0.92$) enables tight focusing of light, creating non-paraxial conditions that facilitate spin-orbit interactions (SOI). This section provides a detailed theoretical explanation of the DFC's role, focusing on how its tight focusing leads to polarization redistribution, geometric phase gradients, and the resulting spin-to-orbital angular momentum (OAM) conversion.

Longitudinal Electric Field Due to Tight Focusing

From Maxwell's equations, the electric field must remain orthogonal to the wavevector $\vec{k}$. In tightly focused beams, the wavevector has both longitudinal (k_z) and transverse components (k_x, k_y). As a result, the electric field, originally confined to the transverse plane in a paraxial beam, develops a longitudinal component E_z . This longitudinal field becomes increasingly significant as NA increases.

- **Low NA regime:** Most rays are close to the optical axis (small angle θ), hence $E_z \approx 0$.
- **High NA regime:** Rays farther from the axis bend significantly (large θ), resulting in appreciable E_z components.

This redistribution of the electric field across E_x , E_y , and E_z modifies the local polarization state across the focal region. What was initially a spatially homogeneous transverse polarization (set by the PSG) becomes inhomogeneous, exhibiting spatial variation in both magnitude and orientation due to this vectorial coupling.

Geometric Phase Gradient Due to Polarization Variation

For an input polarization (e.g., linear along x), the electric field is redistributed into transverse (E_x, E_y) and longitudinal (E_z) components, with the exact distribution depending on θ and ϕ , leading to a spatially inhomogeneous polarization distribution. This redistribution is described by the following focusing matrix:

$$\begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} = \begin{pmatrix} A + B \cos 2\phi & B \sin 2\phi & C \cos \phi \\ B \sin 2\phi & A - B \cos 2\phi & C \sin \phi \\ -C \cos \phi & -C \sin \phi & A + B \end{pmatrix} \begin{pmatrix} E_x^{\text{in}} \\ E_y^{\text{in}} \\ 0 \end{pmatrix}$$

$$\text{where: } A = \frac{1 + \cos \theta}{2}, \quad B = \frac{1 - \cos \theta}{2}, \quad C = \sin \theta$$

For linear polarization along x , the E_z components at the focus are approximately given by:

$$E_z \propto \sin \theta \cos \phi$$

This expression shows that E_z introduces an azimuthal dependence ($\cos \phi$), making the polarization state spatially inhomogeneous across the beam's cross-section.

Now, we know that the polarization state varies azimuthally. Since any change in polarization

leads to the acquisition of a geometric phase (also known as the Pancharatnam–Berry phase), the evolving polarization state acquires such a phase. In this case, the polarization changes from an initial state (set by the PSG) to a final state after passing through the dark-field condenser (DFC). As the final polarization state also varies azimuthally, the acquired geometric phase becomes azimuthally dependent as well. Therefore, a geometric phase gradient is established with respect to the azimuthal angle ϕ .

Spin-to-Orbital Angular Momentum Conversion

In such a configuration, the spatial variation of the polarization gives rise to spin-orbit interaction (SOI) as follows:

- The spin angular momentum (SAM), associated with polarization (e.g., left or right circular), couples to the spatial phase structure.
- The spatially varying geometric phase leads to the generation of intrinsic orbital angular momentum (OAM).

This process is known as spin-to-intrinsic-orbital-angular-momentum conversion and manifests most effectively when the polarization rotation (or Anisotropic axis rotation) is azimuthally dependent. (see supplementary material for more detailed explanation)

However, under typical DFC operation, this conversion is not easily observable, as the geometric phase gradient is relatively small. To enhance the SOI effect, one must amplify the geometric phase gradient, especially with respect to ϕ . This can be achieved by designing the system to support stronger azimuthal variation in polarization, such as through the use of a waveguided plasmonic crystal (WPC).

3.2 Role of WPC in Spin-Orbit Interaction

Waveguided plasmonic crystals (WPCs) are nanostructured systems that integrate a periodic gold (Au) grating on a 190 nm indium tin oxide (ITO) waveguiding layer atop a glass substrate, enabling unique light-matter interactions through the simultaneous excitation of photonic and plasmonic resonances. This section provides a detailed theoretical explanation of the WPC's role in spin-direction-spin coupling, focusing on the excitation of waveguide and plasmonic modes, their polarization-dependent interactions, the role of Fano resonances, and the scattering mechanisms that make these effects observable in the far field.

Excitation of Photonic and Plasmonic Resonances

WPCs support two distinct types of resonances, which are critical to their functionality:

- **Photonic Waveguide Modes:** These are electromagnetic modes confined within the ITO layer due to total internal reflection. The condition for confinement requires the transverse wavevector components (k_x, k_y) to satisfy the waveguide's dispersion relation, determined by the effective refractive index ($n_{\text{eff}} > 1$). The wavevector magnitude of these modes is $|\mathbf{k}_w| \approx n_{\text{eff}}k_0$, where $k_0 = 2\pi/\lambda$ is the free-space wavevector. Since $n_{\text{eff}} > 1$, $|\mathbf{k}_w| > k_0$, meaning these modes are confined unless scattered. (Explained further later in this section)

- **Plasmonic Resonances:** These are collective oscillations of electrons (surface plasmons) at the gold grating's surface, excited when light couples with the metal. Surface plasmon modes are confined, as $|\mathbf{k}_{\text{spr}}| > k_0$, because surface plasmons carry higher momentum than freely propagating light of the same frequency.

The tight focusing of light by the dark-field condenser (DFC) introduces additional transverse momentum components (k_x, k_y), enabling the **excitation** of these leaky quasiguided modes. The DFC's high numerical aperture ($\text{NA} \approx 0.8\text{--}1.2$) ensures that the incident light has sufficient momentum to couple into the waveguide modes, initiating the interaction with the WPC.

Polarization-Dependent Mode Excitation

The polarization state of the incident light, defined relative to the grating axis, determines the efficiency of mode excitation:

- **Transverse Magnetic (TM) Modes:** For TM polarization, the electric field ($\mathbf{E}$) is perpendicular to the grating axis (i.e, along y-axis), aligning parallel to the grating vector ($\mathbf{G} = 2\pi/d$, where d is the grating period). This orientation allows the electric field to couple effectively with the grating, enabling the excitation of surface plasmons. The coupling occurs because the electric field component parallel to $\mathbf{G}$ interacts with the periodic structure, satisfying the momentum-matching condition for plasmon excitation.
- **Transverse Electric (TE) Modes:** For TE polarization, the electric field is parallel to the grating axis (i.e, along x-axis), perpendicular to $\mathbf{G}$. In this case, the electric field does not couple efficiently with the grating, resulting in minimal or no plasmon excitation. TE modes primarily excite photonic waveguide modes within the ITO layer.

Thus, TM-polarized light excites both photonic and plasmonic resonances, while TE-polarized light is limited to photonic modes, leading to polarization-dependent interactions critical for spin-direction-spin coupling.(thus, Au grating axis is Anisotropic axis which is fixed)

Hybrid Modes and Fano Resonance

When the frequencies of photonic waveguide modes and plasmonic resonances overlap (typically around 530–590 nm), they interfere to form hybrid modes. This interference results in **Fano resonances**, characterized by sharp, asymmetric spectral features due to the coupling between a broad continuum state (photonic mode) and a narrow discrete state (plasmonic mode).

Fano resonances significantly enhance the polarization-dependent properties of the WPC, including:

- **Diattenuation:** Differential attenuation of polarization states, where TM-polarized light is preferentially transmitted or reflected compared to TE-polarized light, due to stronger plasmonic coupling.
- **Retardation:** Differential phase shifts between orthogonal polarization components, acting as a retarder and contributing to geometric phase effects.

These enhanced properties amplify the spin-orbit interactions (SOI), making effects like spin-direction-spin coupling more pronounced and observable in the far field.

Scattering and Leaky Modes

The periodic Au grating acts as a diffraction grating, scattering the confined waveguide and plasmonic modes into free space. The scattering process modifies the wavevector of the modes according to:

$$\mathbf{k}_{\text{scatter}} = \mathbf{k}_{\text{mode}} + m\mathbf{G}$$

where m is the diffraction order ($\pm 1, \pm 2, \pm 3, \dots$). For the scattered light to radiate into free space and be detectable in the far field, its wavevector must fall within the light cone, i.e., $|\mathbf{k}_{\text{scatter}}| \leq k_0$. The light cone is defined by:

$$|\mathbf{k}| = \sqrt{k_x^2 + k_y^2 + k_z^2} = k_0$$

In the plane of the WPC (xy-plane), the z-component of the wavevector is:

$$k_z = \sqrt{k_0^2 - (k_x^2 + k_y^2)}$$

- If $k_x^2 + k_y^2 > k_0^2$, k_z becomes imaginary, resulting in an evanescent wave that decays exponentially and remains confined within the plane.
- If $k_x^2 + k_y^2 \leq k_0^2$, k_z is real, allowing the wave to propagate into free space as radiation.

For waveguide and plasmonic modes:

Waveguide Modes: $|\mathbf{k}_w| \approx n_{\text{eff}}k_0 > k_0$, so they are confined unless scattered by the grating into the light cone.

Plasmonic Modes: $|\mathbf{k}_{\text{spr}}| > k_0$, requiring the grating to provide additional momentum for excitation and scattering.

Only scattered waves with $|\mathbf{k}_{\text{scatter}}| \leq k_0$ radiate into free space, forming ring-like patterns in the far field. These rings arise because the scattered light forms a conical radiation pattern, with the angle determined by the transverse wavevector components (k_x, k_y).

However, the microscope objective's limited numerical aperture (NA) restricts the collection of higher-order diffraction modes. For example:

- Modes with $\mathbf{k}_w \pm 2\mathbf{G}, \mathbf{k}_w \pm 3\mathbf{G}, \dots$ often have $|\mathbf{k}_{\text{scatter}}| > k_0 \cdot \text{NA}$, making them undetectable.
- For lower orders (e.g., $\mathbf{k}_w \pm \mathbf{G}$), only arc segments of the diffraction rings may be collected.
- The zeroth-order mode ($m = 0$) has $|\mathbf{k}_w| > k_0$, so it remains confined and does not radiate.

4. Mathematical description

This section derives the Mueller matrix for a linear diattenuating retarder with an azimuthally varying anisotropic axis for the Dark-Field Condenser (DFC) and a fixed anisotropic axis for the Waveguided Plasmonic Crystal (WPC). The derivation starts from fundamental Jones matrices for a linear retarder and diattenuator, combines them, converts to a Mueller matrix, and rotates it to account for orientation. The reasoning for modeling the DFC with an azimuthally varying axis and the WPC with a fixed axis is provided, based on their physical roles in spin-direction-spin coupling.

4.1 Developing Jones Matrix

Jones Matrix for Linear Retarder ($\theta = 0$)

A linear retarder with its fast axis along the x -axis ($\theta = 0$) introduces a phase difference δ between the x - and y -polarized components of the electric field. The Jones matrix is:

$$J_{\text{retarder}} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\delta} \end{pmatrix}$$

Here, δ is the retardance, representing the phase shift between the fast (x) and slow (y) axes.

Jones Matrix for Linear Diattenuator ($\theta = 0$)

A linear diattenuator with principal axes along x and y ($\theta = 0$) has different amplitude transmission coefficients t_x and t_y . The Jones matrix is:

$$J_{\text{diattenuator}} = \begin{pmatrix} t_x & 0 \\ 0 & t_y \end{pmatrix}$$

The diattenuation d is defined as:

$$d = \frac{t_x^2 - t_y^2}{t_x^2 + t_y^2}$$

Combined Jones Matrix for Diattenuating Retarder ($\theta = 0$)

Since both the retarder and diattenuator have their axes aligned at $\theta = 0$, their Jones matrices can be multiplied:

$$J = J_{\text{diattenuator}} \cdot J_{\text{retarder}} = \begin{pmatrix} t_x & 0 \\ 0 & t_y \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & e^{i\delta} \end{pmatrix} = \begin{pmatrix} t_x & 0 \\ 0 & t_y e^{i\delta} \end{pmatrix}$$

This represents a system that applies diattenuation and retardation along the same axes.

4.2 Conversion to Mueller Matrix

The Mueller matrix relates the input and output Stokes vectors, which describe the polarization state. For a Jones matrix $J = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$, where $a = t_x$, $b = t_y e^{i\delta}$, the Mueller matrix is computed using:

$$M = A(J \otimes J^*)A^{-1}$$

where A is the transformation matrix between Stokes and coherency vectors:

$$A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & i & -i & 0 \end{pmatrix}, \quad A^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -i \\ 0 & 0 & 1 & i \\ 1 & -1 & 0 & 0 \end{pmatrix}$$

Compute the Kronecker product $J \otimes J^*$:

$$J = \begin{pmatrix} t_x & 0 \\ 0 & t_y e^{i\delta} \end{pmatrix}, \quad J^* = \begin{pmatrix} t_x & 0 \\ 0 & t_y e^{-i\delta} \end{pmatrix}$$

$$J \otimes J^* = \begin{pmatrix} t_x t_x & t_x \cdot 0 & 0 \cdot t_x & 0 \cdot 0 \\ t_x \cdot 0 & t_x(t_y e^{-i\delta}) & 0 \cdot t_x & 0 \cdot (t_y e^{-i\delta}) \\ 0 \cdot t_x & 0 \cdot (t_y e^{-i\delta}) & (t_y e^{i\delta})t_x & (t_y e^{i\delta}) \cdot 0 \\ 0 \cdot 0 & 0 \cdot (t_y e^{-i\delta}) & (t_y e^{i\delta}) \cdot 0 & (t_y e^{i\delta})(t_y e^{-i\delta}) \end{pmatrix} = \begin{pmatrix} t_x^2 & 0 & 0 & 0 \\ 0 & t_x t_y e^{-i\delta} & 0 & 0 \\ 0 & 0 & t_y t_x e^{i\delta} & 0 \\ 0 & 0 & 0 & t_y^2 \end{pmatrix}$$

Next, compute $M = A(J \otimes J^*)A^{-1}$.

we get,

$$M(0) = \begin{pmatrix} \frac{t_x^2 + t_y^2}{2} & \frac{t_x^2 - t_y^2}{2} & 0 & 0 \\ \frac{t_x^2 - t_y^2}{2} & \frac{t_x^2 + t_y^2}{2} & 0 & 0 \\ 0 & 0 & t_x t_y \cos \delta & t_x t_y \sin \delta \\ 0 & 0 & -t_x t_y \sin \delta & t_x t_y \cos \delta \end{pmatrix}$$

Normalize by setting $\frac{t_x^2 + t_y^2}{2} = 1$, so $d = \frac{t_x^2 - t_y^2}{t_x^2 + t_y^2} = \frac{t_x^2 - t_y^2}{2}$, and $t_x t_y = \sqrt{1 - d^2}$. Thus:

$$M(0, d, \delta) = \begin{pmatrix} 1 & d & 0 & 0 \\ d & 1 & 0 & 0 \\ 0 & 0 & \sqrt{1 - d^2} \cos \delta & \sqrt{1 - d^2} \sin \delta \\ 0 & 0 & -\sqrt{1 - d^2} \sin \delta & \sqrt{1 - d^2} \cos \delta \end{pmatrix}$$

This is the Mueller matrix for $\theta = 0$.

Rotation to General Orientation

To account for an anisotropic axis at angle θ , rotate the Mueller matrix using the rotation matrix:

$$R(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\theta & \sin 2\theta & 0 \\ 0 & -\sin 2\theta & \cos 2\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The rotated Mueller matrix is:

$$M(\theta, d, \delta) = R(-\theta)M(0, d, \delta)R(\theta)$$

The result:

$$M(\theta, d, \delta) = \begin{pmatrix} 1 & d \cos 2\theta & d \sin 2\theta & 0 \\ d \cos 2\theta & \cos^2 2\theta + x \cos \delta \sin^2 2\theta & \cos 2\theta \sin 2\theta (1 - x \cos \delta) & -x \sin \delta \sin 2\theta \\ d \sin 2\theta & \cos 2\theta \sin 2\theta (1 - x \cos \delta) & \sin^2 2\theta + x \cos \delta \cos^2 2\theta & x \sin \delta \cos 2\theta \\ 0 & x \sin \delta \sin 2\theta & -x \sin \delta \cos 2\theta & x \cos \delta \end{pmatrix}$$

where $\sqrt{1 - d^2} = x$.

4.3 DFC Mueller Matrix

For the DFC, the anisotropic axis varies with the azimuthal angle ϕ due to tight focusing, so $\theta = \phi$:

$$M_{\text{DFC}}(\phi, d_f, \delta_f) = \begin{pmatrix} 1 & d_f \cos 2\phi & d_f \sin 2\phi & 0 \\ d_f \cos 2\phi & \cos^2 2\phi + x_f \cos \delta_f \sin^2 2\phi & \cos 2\phi \sin 2\phi (1 - x_f \cos \delta_f) & -x_f \sin \delta_f \sin 2\phi \\ d_f \sin 2\phi & \cos 2\phi \sin 2\phi (1 - x_f \cos \delta_f) & \sin^2 2\phi + x_f \cos \delta_f \cos^2 2\phi & x_f \sin \delta_f \cos 2\phi \\ 0 & x_f \sin \delta_f \sin 2\phi & -x_f \sin \delta_f \cos 2\phi & x_f \cos \delta_f \end{pmatrix}$$

Here, d_f is the focusing diattenuation, δ_f is the focusing retardance and $\sqrt{1 - d_f^2} = x_f$

4.4 WPC Mueller Matrix

For the WPC, the gold grating fixes the anisotropic axis horizontally ($\theta = 0$):

$$M_{\text{WPC}}(0, d_{\text{wpc}}, \delta_{\text{wpc}}) = \begin{pmatrix} 1 & d_{\text{wpc}} & 0 & 0 \\ d_{\text{wpc}} & 1 & 0 & 0 \\ 0 & 0 & x_{\text{wpc}} \cos \delta_{\text{wpc}} & x_{\text{wpc}} \sin \delta_{\text{wpc}} \\ 0 & 0 & -x_{\text{wpc}} \sin \delta_{\text{wpc}} & x_{\text{wpc}} \cos \delta_{\text{wpc}} \end{pmatrix}$$

Here, d_{wpc} and δ_{wpc} are the WPC's diattenuation and retardance.
where, $\sqrt{1 - d_{\text{wpc}}^2} = x_{\text{wpc}}$

4.5 Reasoning for Modeling

- **DFC Modeling:** The DFC's high numerical aperture (0.8–1.2) creates a non-paraxial beam, redistributing the electric field into transverse and longitudinal components, varying with ϕ . This spatial variation induces a geometric phase (Pancharatnam-Berry phase), driving spin-orbit interactions. The Mueller matrix with $\theta = \phi$ models this azimuthal dependence, capturing diattenuation (from differential transmission) and retardation (from phase shifts).
- **WPC Modeling:** The WPC's periodic gold grating fixes the anisotropic axis ($\theta = 0$), leading to consistent polarization effects. Diattenuation arises from preferential transmission of transverse magnetic (TM) polarization, exciting plasmons, while retardation results from phase differences in hybridized modes, enhanced by Fano resonances.

5. Simulation results

5.1 The Plots

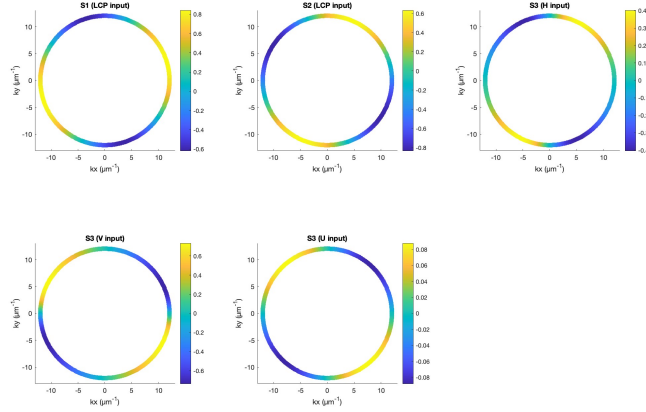


Figure 2: Stokes parameters S_1 , S_2 for LCP input, and S_3 for horizontal, vertical, and unpolarized inputs, showing azimuthal polarization modulation and spin-dependent scattering in the WPC.

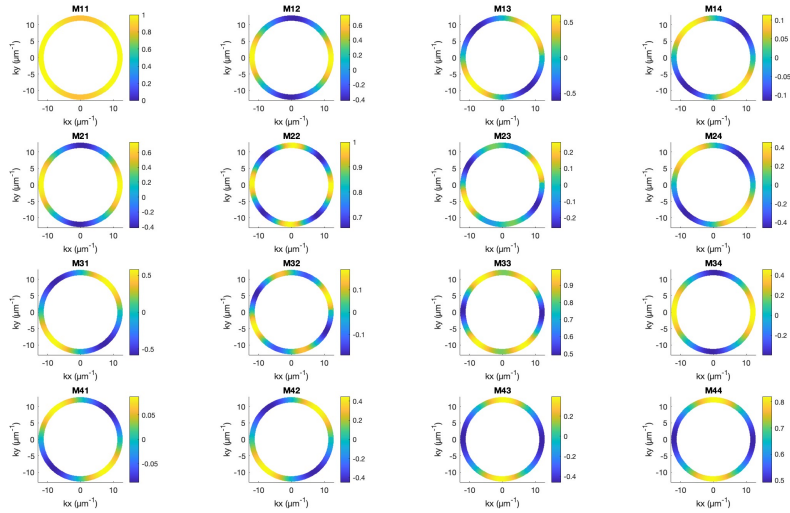


Figure 3: Mueller matrix elements in the Fourier plane showing SOI-induced polarization effects in the WPC. M_{14} and M_{41} indicate forward and inverse SHE, respectively.

5.2 Key Observations and Interpretations

from Figure 2: Stokes Parameters for Various Input Polarizations

- The Stokes parameters S_1 and S_2 for left circularly polarized (LCP) input light exhibit azimuthal variation, indicating polarization redistribution due to tight focusing by the dark-field condenser (DFC). This is attributed to the longitudinal electric field component E_z , introducing spatial inhomogeneity.
- The S_3 component shows ϕ -dependent lobes for horizontally, vertically, and unpolarized inputs, confirming spin-dependent directional guiding (forward SHE) and direction-dependent spin acquisition (inverse SHE).

from Figure 3: Interpretation of each Mueller Matrix elements

The Mueller matrix in the Fourier (k-) domain highlights polarization transformations induced by the WPC.

- M_{11} : Represents total unpolarized scattered intensity.
- M_{14} : Quantifies **forward SHE** through circular diattenuation, showing momentum-dependent intensity differences for RCP vs. LCP inputs, revealing spin-dependent directional excitation of quasiguided modes.
- M_{41} : Quantifies **inverse SHE** via circular diattenuation, showing momentum-dependent circular polarization in scattered light, indicating spin-dependent directional effects.
- M_{22}, M_{33}, M_{44} : Preserve linear polarization along horizontal-vertical (M_{22}), $\pm 45^\circ$ axes (M_{33}), and circular polarization (M_{44}), respectively, while also showing corresponding linear or circular retardance.
- $M_{12}, M_{21}, M_{13}, M_{31}$: Represent linear diattenuation — along horizontal-vertical axes (M_{12}, M_{21}) and $\pm 45^\circ$ axes (M_{13}, M_{31}). Their azimuthal variation indicates the WPC's preference for horizontal polarization due to grating alignment and tight focusing.
- $M_{23}, M_{32}, M_{24}, M_{42}, M_{34}, M_{43}$: Represent polarization coupling terms — M_{23}, M_{32} describe linear retardance via coupling between H/V and $\pm 45^\circ$ polarizations; M_{24}, M_{42} indicate linear-to-circular polarization coupling; and M_{34}, M_{43} reflect circular retardance and linear-circular coupling.

6. Conclusion

This study numerically explored spin-direction-spin coupling in waveguided plasmonic crystals (WPCs), replicating key experimental observations from a dark-field Fourier domain Mueller matrix microscope.

Results confirmed that spin-orbit interaction (SOI), enhanced by Fano resonance, enables spin-dependent guiding (forward SHE) and direction-dependent spin acquisition (inverse SHE). Stokes parameters (S_1 , S_2 , S_3) and Mueller matrix elements M_{14} and M_{41}) revealed azimuthal polarization modulation and directional selectivity.

These findings underscore WPCs' potential in next-generation nanophotonic devices like optical switches and polarization sorters, with applications in optical communication and quantum optics. Future work may apply these methods to other nanostructured platforms to broaden their impact.

Supplementary info.

1. How can refractive index(n) be imaginary?

Step 1: Deriving wave equation for $\vec{E}$ in a conducting medium

Maxwell's equations in free space (no bound charges, but with free current density $\mathbf{J}_f$) in SI units are:

1. $\nabla \cdot \mathbf{E} = 0$ (no free charges in the medium),
2. $\nabla \cdot \mathbf{B} = 0$,
3. $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$,
4. $\nabla \times \mathbf{B} = \mu \mathbf{J}_f + \mu \epsilon \frac{\partial \mathbf{E}}{\partial t}$.

For a conductor, the free current density is given by Ohm's law:

$$\mathbf{J}_f = \sigma \mathbf{E},$$

where σ is the conductivity, μ is the permeability, and ϵ is the permittivity. Take the curl of the third Maxwell equation:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla \times \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = -\frac{\partial}{\partial t} (\nabla \times \mathbf{B}).$$

Using the vector identity $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$, and since $\nabla \cdot \mathbf{E} = 0$, we get:

$$\nabla \times (\nabla \times \mathbf{E}) = -\nabla^2 \mathbf{E}.$$

Substitute $\nabla \times \mathbf{B}$ from the fourth Maxwell equation:

$$\nabla \times \mathbf{B} = \mu \sigma \mathbf{E} + \mu \epsilon \frac{\partial \mathbf{E}}{\partial t}.$$

So:

$$-\frac{\partial}{\partial t} (\nabla \times \mathbf{B}) = -\frac{\partial}{\partial t} \left(\mu \sigma \mathbf{E} + \mu \epsilon \frac{\partial \mathbf{E}}{\partial t} \right) = -\mu \sigma \frac{\partial \mathbf{E}}{\partial t} - \mu \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

Equate both sides:

$$\begin{aligned} -\nabla^2 \mathbf{E} &= -\mu \sigma \frac{\partial \mathbf{E}}{\partial t} - \mu \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}. \\ \nabla^2 \mathbf{E} &= \mu \sigma \frac{\partial \mathbf{E}}{\partial t} + \mu \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}. \end{aligned}$$

This is the wave equation for the electric field in a conducting medium.

Step 2: Deriving the complex form of wavevector

Assume a time-harmonic electric field:

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}(\mathbf{r})e^{-i\omega t}.$$

Compute the time derivatives:

$$\begin{aligned}\frac{\partial \mathbf{E}}{\partial t} &= -i\omega \mathbf{E}e^{-i\omega t} = -i\omega \mathbf{E}, \\ \frac{\partial^2 \mathbf{E}}{\partial t^2} &= (-i\omega)^2 \mathbf{E}e^{-i\omega t} = -\omega^2 \mathbf{E}.\end{aligned}$$

Substitute into the wave equation:

$$\begin{aligned}\nabla^2 \mathbf{E} &= \mu\sigma(-i\omega \mathbf{E}) + \mu\epsilon(-\omega^2 \mathbf{E}). \\ \nabla^2 \mathbf{E} &= -i\omega\mu\sigma \mathbf{E} - \mu\epsilon\omega^2 \mathbf{E}. \\ \nabla^2 \mathbf{E} &= -\mu\omega^2 \left(\epsilon + \frac{i\sigma}{\omega} \right) \mathbf{E}.\end{aligned}$$

Define the effective complex permittivity:

$$\epsilon_{\text{eff}} = \epsilon + \frac{i\sigma}{\omega}.$$

So the wave equation becomes:

$$\nabla^2 \mathbf{E} = -\mu\omega^2 \epsilon_{\text{eff}} \mathbf{E}.$$

Assume a plane wave solution propagating in the z -direction:

$$\mathbf{E} = \mathbf{E}_0 e^{i\mathbf{k} \cdot \mathbf{r}} e^{-i\omega t} = \mathbf{E}_0 e^{ikz} e^{-i\omega t}.$$

Compute the Laplacian:

$$\nabla^2 \mathbf{E} = \frac{\partial^2 \mathbf{E}}{\partial z^2} = (ik)^2 \mathbf{E} = -k^2 \mathbf{E}.$$

Substitute into the wave equation:

$$\begin{aligned}-k^2 \mathbf{E} &= -\mu\omega^2 \epsilon_{\text{eff}} \mathbf{E}. \\ k^2 &= \mu\omega^2 \epsilon_{\text{eff}} = \mu\omega^2 \left(\epsilon + \frac{i\sigma}{\omega} \right).\end{aligned}$$

Express $\epsilon = \epsilon_r \epsilon_0$ and $\mu = \mu_0$, and note that $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$:

$$\begin{aligned}k^2 &= \omega^2 \mu_0 \left(\epsilon_r \epsilon_0 + \frac{i\sigma}{\omega} \right) = \omega^2 \mu_0 \epsilon_0 \left(\epsilon_r + \frac{i\sigma}{\omega \epsilon_0} \right). \\ k^2 &= \frac{\omega^2}{c^2} \left(\epsilon_r + \frac{i\sigma}{\omega \epsilon_0} \right). \\ k &= \frac{\omega}{c} \sqrt{\epsilon_r + \frac{i\sigma}{\omega \epsilon_0}}.\end{aligned}$$

This k is complex, indicating both propagation and attenuation.

Step 3: Relate Wave Vector to Refractive Index

In vacuum, $\epsilon_r = 1$, $\sigma = 0$, the wave vector is:

$$k_0 = \frac{\omega}{c} \sqrt{1 + i.0}$$

$$k_0 = \frac{\omega}{c}.$$

Refractive index is defined as,

$$n = \frac{\text{speed of light in vacuum}}{\text{speed of light in medium}} = \frac{c}{v}$$

wavevector(or wave number), $k = \frac{2\pi}{\lambda} = \frac{2\pi f}{c} = \frac{w}{c} = k_0$

In a medium, phase velocity, $v = f\lambda$

so,

$$k = \frac{2\pi}{\lambda} = \frac{w}{v} = \frac{w}{c/n} = \frac{w}{c} n$$

putting the complex form of k:

$$k = \frac{\omega}{c} \sqrt{\epsilon_r + \frac{i\sigma}{\omega\epsilon_0}} = \frac{\omega}{c} N.$$

$$N = \sqrt{\epsilon_r + \frac{i\sigma}{\omega\epsilon_0}}.$$

where,

$$N = n + in'$$

2. How does n affect phase-shift and Polarisation state?

Step 1: Electric Field with Complex Wave Vector

For a wave propagating in the $+z$ -direction:

$$E(z, t) = E_0 e^{i(kz - \omega t)}.$$

Since k is complex, let:

$$k = k_r + ik_i.$$

$$E(z, t) = E_0 e^{i((k_r + ik_i)z - \omega t)} = E_0 e^{i(k_r z - \omega t)} e^{-k_i z}.$$

- The term $e^{i(k_r z - \omega t)}$ represents the oscillating part. - The term $e^{-k_i z}$ represents exponential decay.

Since $k = \frac{\omega}{c}(n + in')$:

$$k_r = \frac{\omega}{c}n, \quad k_i = \frac{\omega}{c}n'.$$

$$E(z, t) = E_0 e^{i(\frac{\omega}{c}nz - \omega t)} e^{-\frac{\omega}{c}n'z}.$$

The imaginary part n' causes attenuation.

Step 2: Polarization and Phase Shifts

Consider the electric field components:

$$E_x = E_{0x} e^{i(kz - \omega t)} e^{-i\phi_x}, \quad E_y = E_{0y} e^{i(kz - \omega t)} e^{-i\phi_y}.$$

The ratio:

$$\frac{E_y}{E_x} = \frac{E_{0y}}{E_{0x}} e^{-i(\phi_y - \phi_x)} = \frac{E_{0y}}{E_{0x}} e^{-i\delta},$$

where $\delta = \phi_y - \phi_x$. This phase difference affects the polarization state.

Step 3: Birefringence and Dichroism

For a medium with thickness d , the wave experiences different refractive indices:

$$E_x = E_{0x} e^{i\frac{\omega}{c}(n_x + in'_x)z} = E_{0x} e^{i\frac{\omega}{c}n_x z} e^{-\frac{\omega}{c}n'_x z},$$

$$E_y = E_{0y} e^{i\frac{\omega}{c}(n_y + in'_y)z} = E_{0y} e^{i\frac{\omega}{c}n_y z} e^{-\frac{\omega}{c}n'_y z}.$$

After distance d :

$$E_x = E_{0x} e^{i\frac{\omega}{c}n_x d} e^{-\frac{\omega}{c}n'_x d}, \quad E_y = E_{0y} e^{i\frac{\omega}{c}n_y d} e^{-\frac{\omega}{c}n'_y d}.$$

$$\frac{E_y}{E_x} = \frac{E_{0y}}{E_{0x}} e^{i\frac{\omega}{c}(n_y - n_x)d} e^{-\frac{\omega}{c}(n'_y - n'_x)d} = \frac{E_{0y}}{E_{0x}} e^{i\frac{\omega}{c}\Delta n d} e^{-\frac{\omega}{c}\Delta n' d}.$$

Since $k_0 = \frac{\omega}{c}$:

$$\frac{E_y}{E_x} = \frac{E_{0y}}{E_{0x}} e^{ik_0 \Delta n d} e^{-k_0 \Delta n' d}.$$

- **Birefringence:** The phase term $e^{ik_0 \Delta n d}$ introduces:

$$\delta = k_0 \Delta n d = \frac{\omega}{c} \Delta n d.$$

This is the retardance, causing a change in polarization state.

- **Dichroism:** The amplitude term $e^{-k_0 \Delta n' d}$ causes:

$$\delta' = k_0 \Delta n' d.$$

This is diattenuation, affecting the amplitude ratio.

The term $\frac{E_{0y}}{E_{0x}} e^{i\delta} e^{-\delta'}$ shows that: - Birefringence (Δn) causes a phase difference, altering polarization (e.g., linear to elliptical). - Dichroism ($\Delta n'$) causes differential absorption, modifying the polarization.

Final Expressions

- Complex wave vector:

$$k = \frac{\omega}{c} \sqrt{\epsilon_r + \frac{i\sigma}{\omega\epsilon_0}}.$$

- Complex refractive index:

$$N = n + in' = \sqrt{\epsilon_r + \frac{i\sigma}{\omega\epsilon_0}}.$$

- Electric field:

$$E(z, t) = E_0 e^{i(\frac{\omega}{c}nz - \omega t)} e^{-\frac{\omega}{c}n'z}.$$

- Polarization ratio:

$$\frac{E_y}{E_x} = \frac{E_{0y}}{E_{0x}} e^{ik_0\Delta nd} e^{-k_0\Delta n'd}.$$

- Retardance (birefringence):

$$\delta = k_0\Delta nd.$$

- Diattenuation (dichroism):

$$\delta' = k_0\Delta n'd.$$

This derivation shows how the imaginary part of the refractive index leads to attenuation and how birefringence and dichroism cause phase shifts and amplitude changes, resulting in polarization state changes.

Code

```

1 % Simulates spin to orbital angular momentum conversion in k-space Stokes
  vectors and Mueller matrix elements
2 clear all; close all;
3
4 % Constants
5 NA = 0.86; % Numerical aperture of the microscope objective
6 lambda_nm = 450; % Wavelength in nm
7 k0 = (2 * pi / (lambda_nm * 1e-9)); % Free-space wavevector (m^-1)
8 k_max = k0 * NA; % Maximum wavevector in momentum space
9
10 theta = asin(NA); % Focusing angle corresponding to NA
11 phi = linspace(0, 2 * pi, 500); % Azimuthal angle (full circle)
12
13 d_f = 0.6; % Linear diattenuation due to focusing
14 delta_f = 0.6; % Linear retardance due to focusing
15 d_wpc = 0.25; % Linear diattenuation of WPC
16 delta_wpc = 0.15; % Linear retardance of WPC
17
18 % Mueller matrix for tight focusing (azimuthal linear diattenuating retarder)
19 function M = mueller_focusing(phi, d_f, delta_f)
20     x_f = sqrt(1 - d_f^2);
21     M = [1, d_f * cos(2 * phi), d_f * sin(2 * phi), 0;
22          d_f * cos(2 * phi), cos(2 * phi).^2 + x_f * sin(2 * phi).^2 * cos(
23              delta_f), (1 - x_f * cos(delta_f)) * cos(2 * phi) * sin(2 * phi),
24              -x_f * sin(delta_f) * sin(2 * phi);
25          d_f * sin(2 * phi), (1 - x_f * cos(delta_f)) * cos(2 * phi) * sin(2 *
26              phi), sin(2 * phi).^2 + x_f * cos(2 * phi).^2 * cos(delta_f), x_f
27              * sin(delta_f) * cos(2 * phi);
28          0, x_f * sin(delta_f) * sin(2 * phi), -x_f * sin(delta_f) * cos(2 *
29              phi), x_f * cos(delta_f)];
30 end
31
32 % Mueller matrix for WPC (linear diattenuating retarder with fixed axis)
33 function M = mueller_wpc(d_wpc, delta_wpc)
34     x_wpc = sqrt(1 - d_wpc^2);
35     M = [1, d_wpc, 0, 0;
36          d_wpc, 1, 0, 0;
37          0, 0, x_wpc * cos(delta_wpc), x_wpc * sin(delta_wpc);
38          0, 0, -x_wpc * sin(delta_wpc), x_wpc * cos(delta_wpc)];
39 end
40
41 % Input Stokes vectors
42 S_LCP = [1; 0; 0; -1]; % Left circularly polarized
43 S_H = [1; 1; 0; 0]; % Horizontal linear polarization
44 S_V = [1; -1; 0; 0]; % Vertical linear polarization
45 S_U = [1; 0; 0; 0]; % Unpolarized
46
47 % Initialize arrays for Stokes vectors and Mueller matrix elements
48 S1_LCP = zeros(size(phi));
49 S2_LCP = zeros(size(phi));
50 S3_H = zeros(size(phi));
51 S3_V = zeros(size(phi));
52 S3_U = zeros(size(phi));
53 M_elements = struct();
54 M11_raw = zeros(size(phi)); % Store raw M11 before normalization

```



```

50 for i = 1:4
51     for j = 1:4
52         M_elements.(['M', num2str(i), num2str(j)]) = zeros(size(phi));
53     end
54 end
55
56 % Compute Mueller matrix and Stokes vectors for each azimuthal angle
57 for i = 1:length(phi)
58     M_f = mueller_focusing(phi(i), d_f, delta_f);
59     M_wpc = mueller_wpc(d_wpc, delta_wpc);
60     M = M_wpc * M_f; % Combined Mueller matrix
61
62     % Store raw M11 before normalization
63     M11_raw(i) = M(1,1);
64
65     % Normalize the Mueller matrix to ensure M00 = 1
66     if M(1,1) ~= 0
67         M = M / M(1,1);
68     end
69
70     % Compute output Stokes vectors
71     S_out_LCP = M * S_LCP;
72     S_out_LCP = S_out_LCP / S_out_LCP(1); % Normalize by S0
73     S1_LCP(i) = S_out_LCP(2);
74     S2_LCP(i) = S_out_LCP(3);
75
76     S_out_H = M * S_H;
77     S_out_H = S_out_H / S_out_H(1);
78     S3_H(i) = S_out_H(4);
79
80     S_out_V = M * S_V;
81     S_out_V = S_out_V / S_out_V(1);
82     S3_V(i) = S_out_V(4);
83
84     S_out_U = M * S_U;
85     S_out_U = S_out_U / S_out_U(1);
86     S3_U(i) = S_out_U(4);
87
88     % Store Mueller matrix elements
89     for j = 1:4
90         for k = 1:4
91             M_elements.(['M', num2str(j), num2str(k)])(i) = M(j, k);
92         end
93     end
94 end
95
96 % Simulate k-space arc segments
97 kx = k_max * cos(phi) * 1e-6; % Convert to  $m^{-1}$ 
98 ky = k_max * sin(phi) * 1e-6;
99
100 % Plotting Stokes vectors
101 figure('Position', [100, 100, 1200, 800]);
102 subplot(2, 3, 1);
103 scatter(kx, ky, 50, S1_LCP, 'filled');
104 colormap('parula');
105 colorbar;
106 title('S1 (LCP input)');
107 xlabel('kx (  $m^{-1}$ )');

```

```

108 ylabel('ky_⊓( m ^{-1})');
109 axis equal;
110 xlim([min(kx) - 1, max(kx) + 1]);
111 ylim([min(ky) - 1, max(ky) + 1]);
112
113 subplot(2, 3, 2);
114 scatter(kx, ky, 50, S2_LCP, 'filled');
115 colormap('parula');
116 colorbar;
117 title('S2_⊓(LCP_⊓input)');
118 xlabel('kx_⊓( m ^{-1})');
119 ylabel('ky_⊓( m ^{-1})');
120 axis equal;
121 xlim([min(kx) - 1, max(kx) + 1]);
122 ylim([min(ky) - 1, max(ky) + 1]);
123
124 subplot(2, 3, 3);
125 scatter(kx, ky, 50, S3_H, 'filled');
126 colormap('parula');
127 colorbar;
128 title('S3_⊓(H_⊓input)');
129 xlabel('kx_⊓( m ^{-1})');
130 ylabel('ky_⊓( m ^{-1})');
131 axis equal;
132 xlim([min(kx) - 1, max(kx) + 1]);
133 ylim([min(ky) - 1, max(ky) + 1]);
134
135 subplot(2, 3, 4);
136 scatter(kx, ky, 50, S3_V, 'filled');
137 colormap('parula');
138 colorbar;
139 title('S3_⊓(V_⊓input)');
140 xlabel('kx_⊓( m ^{-1})');
141 ylabel('ky_⊓( m ^{-1})');
142 axis equal;
143 xlim([min(kx) - 1, max(kx) + 1]);
144 ylim([min(ky) - 1, max(ky) + 1]);
145
146 subplot(2, 3, 5);
147 scatter(kx, ky, 50, S3_U, 'filled');
148 colormap('parula');
149 colorbar;
150 title('S3_⊓(U_⊓input)');
151 xlabel('kx_⊓( m ^{-1})');
152 ylabel('ky_⊓( m ^{-1})');
153 axis equal;
154 xlim([min(kx) - 1, max(kx) + 1]);
155 ylim([min(ky) - 1, max(ky) + 1]);
156
157 % Plot Mueller matrix elements
158 figure('Position', [100, 100, 1200, 1200]);
159 t = tiledlayout(4, 4, 'TileSpacing', 'compact');
160 for i = 1:4
161     for j = 1:4
162         nexttile;
163         if i == 1 && j == 1
164             % Plot raw M11 with color scale between 0 and 1
165             scatter(kx, ky, 50, M11_raw, 'filled');

```

```

166         colormap('parula');
167         colorbar;
168         caxis([0 1]); % Set color scale between 0 and 1 as requested
169     else
170         % Plot other normalized Mueller matrix elements
171         scatter(kx, ky, 50, M_elements.(['M', num2str(i), num2str(j)]), '
            filled');
172         colormap('parula');
173         colorbar;
174     end
175     title(['M', num2str(i), num2str(j)]);
176     xlabel('kx_ ( m ^{-1})');
177     ylabel('ky_ ( m ^{-1})');
178     axis equal;
179     xlim([min(kx) - 1, max(kx) + 1]);
180     ylim([min(ky) - 1, max(ky) + 1]);
181 end
182 end

```

Listing 1: Simulation of spin-orbit interactions and Mueller matrix elements in WPC under tight focusing

References

1. **Main Paper:**

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Phys. Rev. Lett. **131**, 193803 (2023).

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2. **Supplemental Material I:**

Detailed theoretical analysis and spectral responses of the waveguided plasmonic crystal.

Available at: <http://link.aps.org/supplemental/10.1103/PhysRevLett.131.193803>

3. **Supplemental Material II:**

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— The End —